

Fir dreptunghiular delimitat de bariere de potential infinite

ORIGIN := 1

$$h := 6.625 \cdot 10^{-34} \quad m_e := 9.1 \cdot 10^{-31} \quad e_e := 1.6 \cdot 10^{-19}$$

$$\hbar := \frac{h}{2 \cdot \pi} \quad m := 0.067 \cdot m_e$$

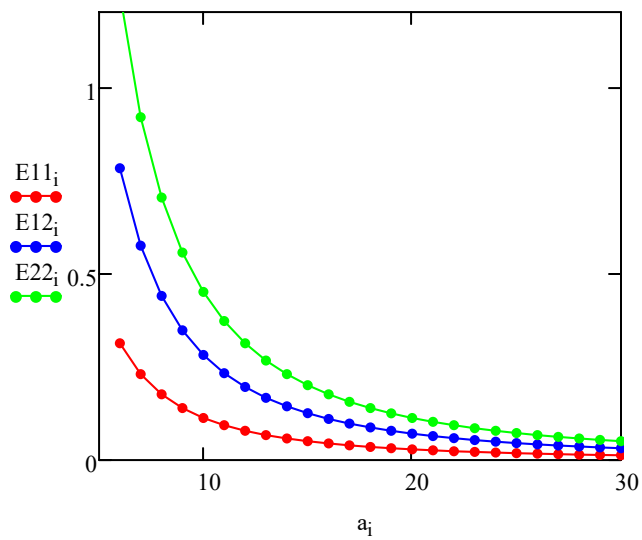
$$i := 1 \dots 26$$

$$a_i := 5 + i \quad \text{latura sectiunii in nanometri}$$

$$E11_i := \frac{\hbar^2 \cdot \pi^2}{2 \cdot m \cdot (a_i)^2 \cdot 10^{-18} \cdot e_e} \cdot (1^2 + 1^2)$$

$$E12_i := \frac{\hbar^2 \cdot \pi^2}{2 \cdot m \cdot (a_i)^2 \cdot 10^{-18} \cdot e_e} \cdot (1^2 + 2^2)$$

$$E22_i := \frac{\hbar^2 \cdot \pi^2}{2 \cdot m \cdot (a_i)^2 \cdot 10^{-18} \cdot e_e} \cdot (2^2 + 2^2)$$



Probabilitatea de localizare a electronului in sectiunea transversala a firului

$$\psi_y(y, n) := \sqrt{\frac{2}{a_5}} \cdot \sin\left(\frac{n \cdot \pi}{a_5} \cdot y\right)$$

$$\psi_z(z, n) := \sqrt{\frac{2}{a_5}} \cdot \sin\left(\frac{n \cdot \pi}{a_5} \cdot z\right)$$

$$ky := 1 \dots 101$$

$$kz := 1 \dots 101$$

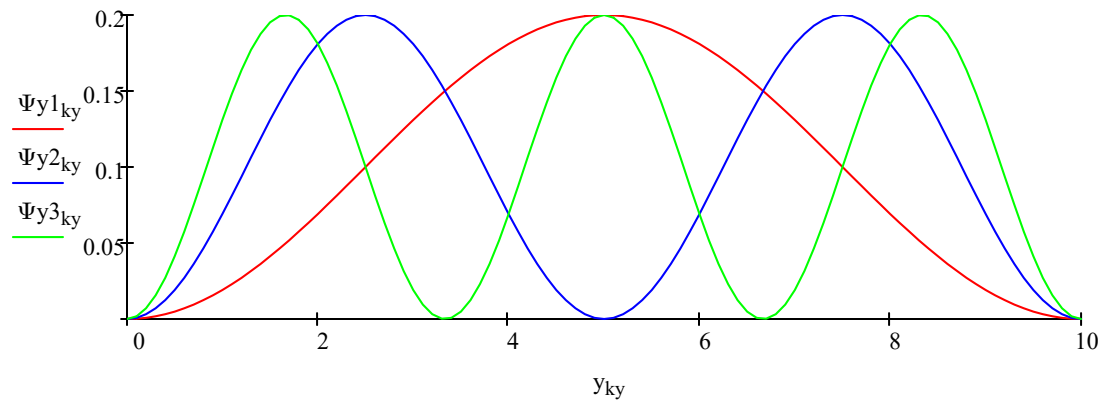
$$y_{ky} := \frac{ky - 1}{10}$$

$$z_{kz} := \frac{kz - 1}{10}$$

$$\Psi_{y1_{ky}} := \left(\left| \psi_y(y_{ky}, 1) \right| \right)^2$$

$$\Psi_{y2_{ky}} := \left(\left| \psi_y(y_{ky}, 2) \right| \right)^2$$

$$\Psi_{y3_{ky}} := \left(\left| \psi_y(y_{ky}, 3) \right| \right)^2$$

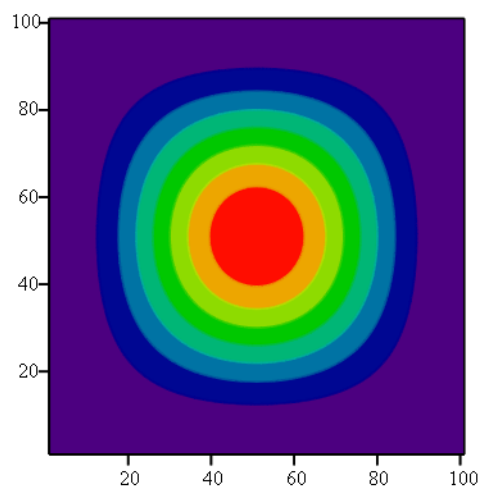


$$\Psi_{11_{ky,kz}} := \left(\left| \psi_y(y_{ky}, 1) \cdot \psi_z(y_{kz}, 1) \right| \right)^2$$

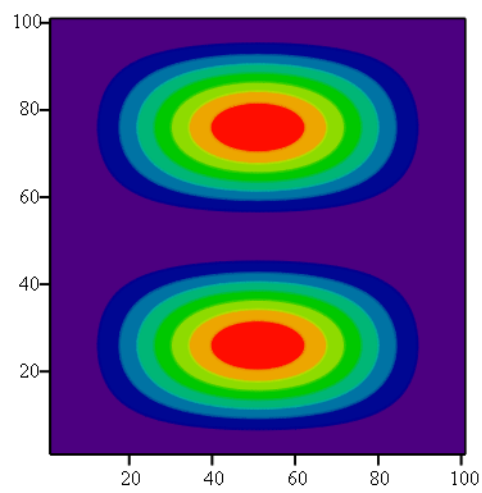
$$\Psi_{12_{ky,kz}} := \left(\left| \psi_y(y_{ky}, 1) \cdot \psi_z(y_{kz}, 2) \right| \right)^2$$

$$\Psi_{21_{ky,kz}} := \left(\left| \psi_y(y_{ky}, 2) \cdot \psi_z(y_{kz}, 1) \right| \right)^2$$

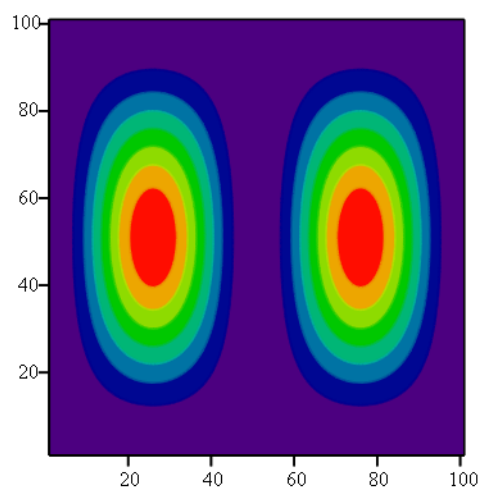
$$\Psi_{22_{ky,kz}} := \left(\left| \psi_y(y_{ky}, 2) \cdot \psi_z(y_{kz}, 2) \right| \right)^2$$



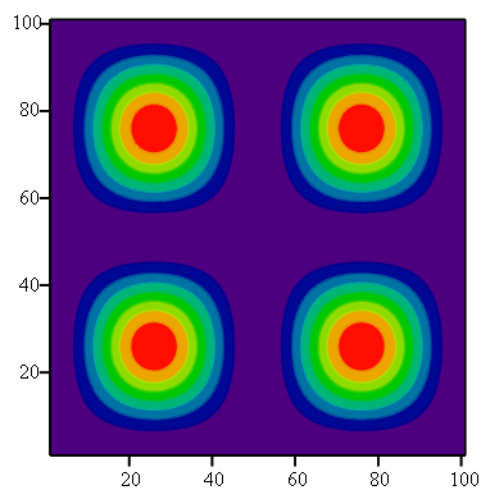
Ψ_{11}



Ψ_{12}



Ψ_{21}



Ψ_{22}